

Project Title: EchoSync :AI-Assisted Pre-Arrival Intelligence & First Response for Unwitnessed Medical Emergencies

Team Name: EchoSync

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Optional supplementary material: <https://canva.link/3jphq3scqtzqxrj>

Executive Summary

Singapore's emergency response ecosystem is highly effective once a distress call is made. However, a critical operational gap remains before help is activated. In unwitnessed home emergencies, particularly among elderly residents living alone, there may be no nearby bystander, no manual activation of an emergency button, and no timely 995 call. This delay significantly affects survival outcomes during time-sensitive medical emergencies such as cardiac arrest, falls, or severe medical distress.

Singapore's ageing population has intensified this challenge. According to national population projections, one in four Singaporeans will be aged 65 and above by 2030. At the same time, reports by CNA and parliamentary responses from the Ministry of Social and Family Development (MSF) have highlighted increasing cases of seniors dying alone at home, with some remaining undiscovered for days. These incidents demonstrate the limitations of current emergency activation systems that rely heavily on conscious manual intervention or neighbour discovery.

EchoSync is a privacy-first, AI-assisted emergency alert and pre-arrival intelligence system designed to identify possible unwitnessed medical emergencies in HDB homes and support faster intervention. The system uses edge-based multi-modal sensing and confidence-scored AI analysis to detect correlated anomalies associated with possible emergencies without relying on invasive surveillance or storing private conversations.

Rather than functioning as an autonomous medical diagnosis platform, EchoSync acts as an intelligence-assisted escalation and responder coordination system. When a high-confidence anomaly is detected, alerts are surfaced through an SCDF-style dispatcher dashboard and can trigger a myResponder-aligned Community First Responder (CFR) workflow to support earlier CPR and AED-assisted intervention before ambulance arrival.

The proposed hackathon MVP focuses on realistic and achievable features within competition scope, including simulated anomaly detection, AI-assisted confidence scoring, dispatcher dashboard integration, responder routing, and simulated AED coordination workflows.

EchoSync is designed to augment, not replace, existing SCDF operations. Through phased deployment, human oversight, and privacy-preserving architecture, the system offers a realistic pathway to improve pre-arrival situational awareness, reduce emergency detection delays, and strengthen community-assisted response during the most critical minutes before SCDF arrival.

1. Proposed Innovative Idea

1.1 Problem Definition and Operational Context

Singapore's emergency response capabilities are highly effective once an incident has been reported. However, a critical operational blind spot remains before emergency activation occurs.

In unwitnessed home emergencies involving elderly residents living alone, there may be no one nearby to witness the incident or trigger a timely emergency response. Current systems such as emergency buttons, panic pull-cords, and wearable devices still depend heavily on conscious manual activation or user compliance. If a resident is unconscious, immobile, or unable to react, emergency response may be significantly delayed.

This issue has become increasingly relevant as Singapore's ageing population continues to grow. National projections estimate that one in four Singaporeans will be aged 65 or above by 2030. In addition, CNA's "Dead Alone in Singapore" documentary series highlighted growing concerns regarding elderly residents whose deaths remained undiscovered for extended periods due to isolation and lack of timely intervention.

From an operational perspective, delayed activation creates several challenges for SCDF responders:

Firstly, response initiation may depend entirely on neighbour discovery or delayed reporting. Secondly, responders may arrive with limited situational awareness regarding what occurred inside the unit. Finally, survival outcomes for incidents such as out-of-hospital cardiac arrest (OHCA) are highly dependent on rapid CPR and AED intervention within the first few minutes.

Existing solutions do not fully address this gap. Wearables require consistent charging and compliance, emergency buttons require conscious activation, and home surveillance cameras create significant privacy concerns in residential environments. While myResponder is highly effective once an alert exists, it cannot address emergencies where no alert is ever triggered.

This creates a strong need for a privacy-preserving, AI-assisted emergency detection and escalation system capable of supporting earlier intervention while remaining operationally realistic for Singapore's public housing environment.

1.2 The EchoSync Solution

EchoSync is a privacy-preserving, AI-assisted emergency alert and pre-arrival intelligence system designed to identify possible unwitnessed medical emergencies in HDB homes.

The system operates as an intelligent enhancement to existing emergency assistance points and responder coordination workflows. Instead of relying solely on manual activation, EchoSync uses edge-based multi-modal sensing to identify correlated anomalies associated with possible medical emergencies.

These anomalies may include prolonged immobility, abrupt motion interruption, abnormal low-resolution thermal patterns, inactivity irregularities, and non-speech acoustic anomalies such as heavy impacts or distress sounds.

To minimise privacy concerns, EchoSync does not use invasive surveillance methods. No raw video, facial recognition data, or private conversation recordings are stored or transmitted. Instead, signals are processed locally using edge AI, and only anonymised data together with explainable alert summaries are surfaced to the dispatcher dashboard.

The system is intentionally designed as a decision-support and escalation platform rather than an autonomous medical diagnosis system. Human operators remain fully involved in verification and escalation workflows.

1.3 Core Operational Workflow

The proposed operational workflow is designed to integrate realistically with existing emergency response ecosystems.

A passive sensing node installed within an opt-in household continuously monitors for correlated emergency anomalies. When the edge AI system detects a high-confidence anomaly, it assigns a confidence score and generates an explainable alert summary.

The alert is then securely transmitted to an SCDF-style dispatcher dashboard for review. If assessed as credible, nearby Community First Responders can be notified through a myResponder-aligned workflow with unit-level incident information and responder guidance.

This allows earlier CPR staging, AED coordination, and pre-arrival situational awareness before ambulance arrival.

The operational objective is not to replace SCDF workflows, but to reduce the dangerous period where no alert has yet been raised despite an emergency occurring.

1.4 Innovation and Strategic Value

EchoSync introduces several innovations aligned with SCDF's operational transformation goals.

Firstly, the system combines privacy-preserving edge sensing with AI-assisted anomaly fusion without relying on invasive home surveillance. Secondly, EchoSync applies explainable confidence scoring rather than binary trigger systems, reducing unnecessary false escalations. Thirdly, the platform supports earlier community-assisted intervention by integrating with responder coordination logic similar to myResponder workflows.

Unlike conventional emergency systems that rely on conscious manual activation, EchoSync enables intelligence-assisted pre-arrival intervention even in unwitnessed emergencies.

This shifts emergency response from purely reactive reporting toward earlier operational awareness and faster responder coordination.

2. MVP Design and Technical Approach

2.1 MVP Scope

The proposed hackathon MVP is intentionally scoped as a realistic software-first operational prototype.

Rather than attempting full autonomous deployment, the MVP focuses on demonstrating the end-to-end workflow from anomaly detection to responder coordination.

The MVP includes simulated multi-modal anomaly detection, AI-assisted confidence scoring, explainable alert reasoning, dispatcher dashboard integration, CFR notification workflows aligned to myResponder logic, simulated AED coordination, incident timeline tracking, and basic node health visibility.

The MVP excludes autonomous medical diagnosis, forced-entry systems, robotics, full estate-wide deployment, or continuous surveillance capabilities. This ensures the prototype remains practical, achievable, and deployment-oriented within hackathon constraints.

2.2 Effective Use of AI

EchoSync uses AI to fuse multiple weak environmental signals into a higher-confidence operational alert.

Instead of relying on a single trigger source, the system combines thermal irregularities, motion inactivity, acoustic anomalies, and behavioural deviations to improve reliability while reducing false positives.

The platform applies explainable AI principles by surfacing operator-readable reasoning such as “impact anomaly + prolonged immobility detected.” This improves dispatcher interpretability and operational trust.

Edge processing is used to minimise privacy risks, reduce bandwidth dependency, and ensure sensitive data remains localised within the household environment.

Confidence-based alert prioritisation enables anomalies to be categorised into low, medium, or high-confidence events. Lower-confidence alerts may be routed for verification instead of immediate escalation.

Importantly, EchoSync adopts a human-in-the-loop operational model where SCDF operators retain full authority over verification and escalation decisions.

2.3 Representative Emergency Workflow

A representative emergency workflow demonstrates the intended operational value of the system.

At T+00 seconds, an elderly resident collapses alone inside an opt-in household. Within approximately 45 seconds, the edge AI system identifies correlated anomalies associated with a possible emergency.

At approximately T+50 seconds, the dispatcher dashboard receives a high-confidence alert and nearby Community First Responders are notified. AED coordination and responder guidance workflows begin shortly after.

By approximately T+2 minutes, nearby CFRs may begin CPR assistance before SCDF responders arrive.

The core operational benefit lies in reducing detection-to-alert delays and improving pre-arrival situational awareness during the critical early minutes of an emergency.

3. Deployment Practicality and Pilot Feasibility

3.1 Deployment Suitability for Singapore

EchoSync is designed specifically for Singapore’s public housing and emergency response context, where privacy, operational practicality, and scalability are essential.

HDB estates provide standardised residential layouts, dense resident populations, existing community responder ecosystems, and established coordination structures involving SCDF, HDB, Town Councils, and healthcare agencies.

These characteristics make Singapore's public housing environment highly suitable for supervised pilot deployment.

3.2 Proposed Pilot Model

A realistic initial pilot should remain limited, supervised, and opt-in.

The recommended pilot scope involves one to two HDB blocks located within senior-heavy precincts such as Silver Zones.

Each participating unit would contain a ceiling-mounted sensing node with local edge processing capabilities. Alerts generated by the system would remain human-reviewed rather than autonomously escalated.

The pilot deployment would include:

- opt-in sensing nodes within selected HDB units,
- enhanced emergency assistance points for manual escalation,
- simulated AED coordination points in common areas,
- secure edge gateways for connectivity support,
- and a dispatcher dashboard for confidence-rated alert management.

The pilot would function as a proof-of-operations model to evaluate whether EchoSync can reduce detection-to-alert time while maintaining operationally acceptable false-alert rates.

3.3 Operational Safeguards and Risk Management

Operational safeguards are critical for deployment realism and public trust.

EchoSync adopts a privacy-first architecture where no raw video, facial recognition data, or private conversations are stored. Participation remains fully opt-in, and only anonymised data is processed.

SCDF operators retain full override authority throughout the workflow. Lower-confidence anomalies can be routed for verification rather than immediate escalation.

The system prioritises verification and operational trust through explainable confidence scoring and human-reviewed escalation workflows.

The platform also incorporates encrypted alert transmission, role-based dashboard access, secure device authentication, fallback connectivity support, and graceful degradation if one sensing source fails.

Importantly, EchoSync does not assume autonomous entry or forced access into residential units. The system's operational value remains in earlier anomaly detection, improved situational awareness, faster responder staging, and accelerated AED coordination.

3.4 Strategic Improvement Over Existing Systems

EchoSync complements existing emergency response systems by addressing their operational blind spots.

Compared with wearables, EchoSync does not depend on charging compliance or user behaviour. Compared with emergency buttons, the system can still function when the resident is unconscious or unable to react.

Compared with myResponder alone, EchoSync helps generate alerts in situations where no bystander is present. Compared with residential surveillance cameras, the platform preserves privacy and is likely to achieve stronger public acceptance.

This positions EchoSync as a practical enhancement to Singapore's existing emergency response ecosystem rather than a replacement for current infrastructure.

4. Expected Impact and Conclusion

4.1 Expected Operational Impact

EchoSync is designed to strengthen the pre-arrival phase of emergency response, particularly during unwitnessed residential emergencies.

By reducing the delay between incident occurrence and emergency activation, the system may improve survival outcomes during time-sensitive incidents such as cardiac arrest and severe falls.

Potential operational benefits include reduced time-to-first-intervention, earlier CFR-assisted CPR before ambulance arrival, faster AED coordination, improved dispatcher situational awareness, and reduced responder uncertainty during entry and assessment.

Confidence-based verification may also reduce unnecessary dispatch burden caused by unreliable single-trigger alert systems.

4.2 Proposed Pilot KPIs

Pilot evaluation can be measured using practical operational indicators.

These include:

- median time from anomaly detection to dashboard alert below 60 seconds,
- CFR acceptance rates within the pilot zone,
- reduction in operator-dismissed false alerts during calibration,
- and measurable reductions in time-to-first-responder action compared with baseline workflows.

These indicators are operationally relevant and aligned with evaluating whether EchoSync meaningfully improves the pre-arrival response window.

4.3 Conclusion

EchoSync supports Singapore's shift toward smarter, more resilient emergency response systems through practical, privacy-conscious innovation.

Rather than replacing frontline responders, the platform is designed to augment SCDF workflows by improving early situational awareness, enabling faster community-assisted intervention, and reducing uncertainty before responder arrival.

The proposal is intentionally scoped to remain realistic, software-first, scalable, and pilot-ready within Singapore's operational and public housing constraints.

Most importantly, EchoSync addresses one of the most dangerous gaps in residential emergency response: the moment when help is needed, but no alert has yet been raised.

References

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